

DeepSeek-V4 Agent Interfaces and DeepSeek Harness

Deep Dive into DeepSeek — Chapter 14

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Foundation Books Project

The one rule of this chapter

A checkpoint can *propose* an action.

It can **not** authorize an effect or own recoverable state.

- DeepSeek-V4 serializes tool calls, preserves tool-conditioned reasoning, and makes Quick Instruction decisions.
- An application-side control loop — DeepSeek Harness — validates, authorizes, confines, executes, and makes outcomes durable.
- One user-visible turn crosses five owners: checkpoint, adapter, application, environment, and evaluation harness.

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Goal: follow one repository-repair turn end to end, and keep every claim attached to the owner that actually establishes it.

1. One turn, five owners

Roadmap

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2. Model-native protocol

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4. Guarded execution, recovery, operating points

One turn, five owners

The running turn: repair a parser regression

checkpoint
(V4)

- The model emits reasoning state plus a call proposal. The adapter parses it into a typed call and *does not execute it*.
- The harness validates policy and capability, executes what it admits, and appends the typed result before the next request is built.

The running turn: repair a parser regression



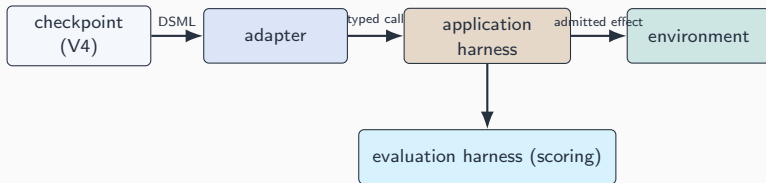
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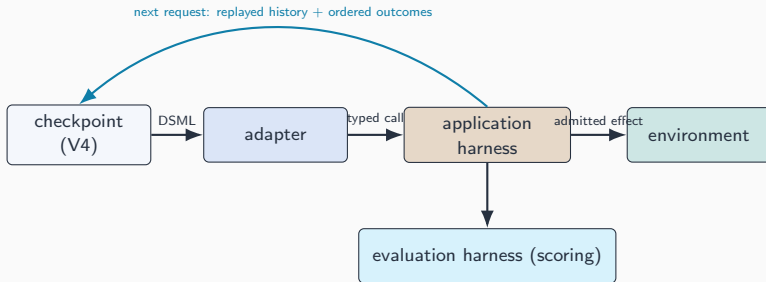
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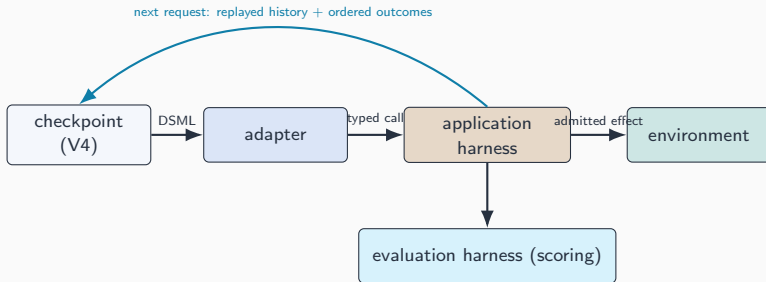
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The chapter's central contract is ownership, not mechanism.

Notation carried through the chapter

Symbol	Meaning in this chapter
U_t, A_t, R_t	user messages, observable calls and results, and returned reasoning records through turn t
$C = (c_1, \dots, c_k)$	assistant-proposed calls in stable call order
$O = (O_1, \dots, O_k)$	ordered durable outcomes, one per proposed call
$E_{\leq t}$	append-only durable session events through step t
D, M_t	session projection and model-visible history $M_t = D(E_{\leq t})$

Two later symbols extend the request side: H_t , the folded request header, and P_r , the deterministic projection for model route r , giving $Q_{t,r} = P_r(M_t, H_t, A)$ over an immutable attachment store A .

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Each symbol names something an owner is responsible for — which is why the chapter can say where a failure came from.

Model-native protocol

From concept to code: the DSML envelope

The report's serialization is a DSML-delimited XML sequence: a `tool_calls` envelope, one or more `invoke` elements, named `parameter` elements, and a marker for literal-string payloads.

from DeepSeek-V4-Pro/DeepSeek_V4.pdf (report grammar)

```
<|DSML|tool_calls>
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A hosted SDK may instead expose an adapter-specific `tool_calls` object with JSON arguments. The application still owns tool-name, schema, permission, result, and recovery checks as distinct steps. [REPORTED: DeepSeek-V4 technical report]

Reasoning replay is conditional on the path

Invariant (V4 tool-conditioned reasoning replay)

On the native tool-calling path, the next request retains $(U_{\leq t}, A_{\leq t}, R_{\leq t})$. On the ordinary no-tool conversational path, the next user turn retains the observable dialogue and starts fresh reasoning.

- V3.2 kept reasoning across tool-result rounds but discarded it at a new user message; V4 keeps it across user-message boundaries too. [REPORTED: DeepSeek-V4 technical report]
- Under the dated Chat Completions adapter, every returned `reasoning_content` value from a tool-call turn must be passed back in all later requests. [REPORTED: official thinking-mode guide]

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R_t is state the client supplies again — not private memory recovered from the checkpoint.

Quick Instruction: one prefix, six auxiliary decisions

Special token

Auxiliary result requested from the shared prefix

`<|action|>`

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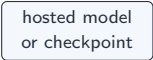
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[REPORTED: DeepSeek-V4 technical report]

Quick Instruction removes duplicated routing prefill. It does not execute the search, validate the source, or transfer effect ownership away from the application.

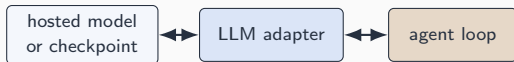
Harness composition and durable state

A plugin tree around a durable log

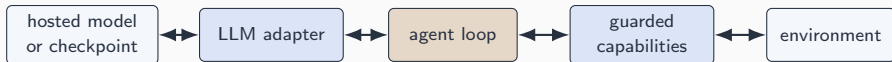


hosted model
or checkpoint

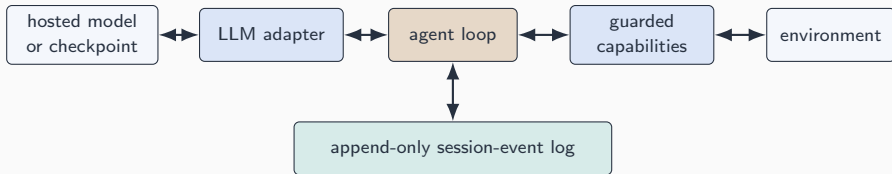
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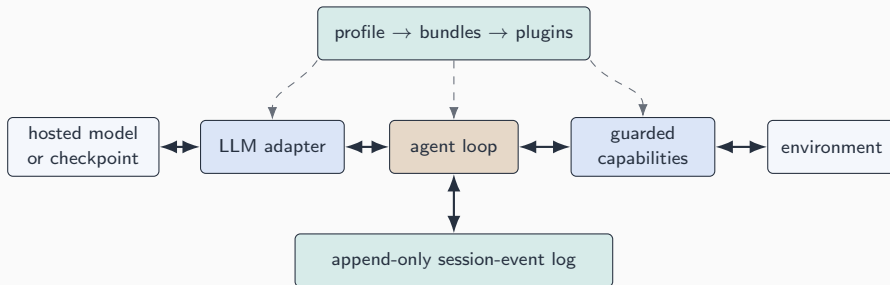
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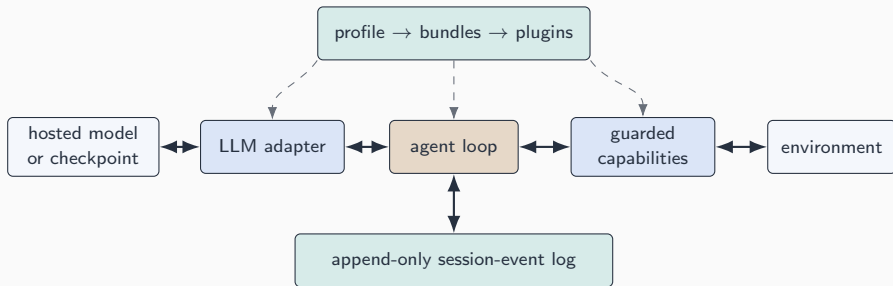


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Consumers depend on a service definition, not a provider, so a model adapter, tool provider, persistence layer, or loop policy can be replaced without moving that responsibility into the checkpoint. [INSPECTED: pinned dsh-v0.1.1-rc.2 snapshot]

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Two clocks: turns and steps

Step

- one model request, plus the tool work its response causes
- ends when its outcomes are durable

Turn

- zero or more steps
- ends when the loop owes no further action

One repair turn, three steps:

1. model call → read tool → append observation

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The clocks can disagree. If the application commits a call proposal as though its effect had happened, a restart can replay a fictitious write. Finalize the model event, admit and execute, *then* commit the result.

What the next request is allowed to be made of

$$M_t = D(E_{\leq t})$$

$$Q_{t,r} = P_r(M_t, H_t, A)$$

Layer	Persistence	Responsibility
session event	durable	message structure plus a stable attachment identifier
normalized attachment	durable	verified, provider-independent content
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Compaction appends a replacement surface citing the range it shadows; it never rewrites old events. The audit log survives the shorter projection. [INSPECTED: pinned dsh-v0.1.1-rc.2 snapshot]

Guarded execution, recovery, operating points

Body timing may race; history may not

Invariant (Ordered tool-result commitment)

For assistant calls $C = (c_1, \dots, c_k)$, durable outcomes appear in the same order $O = (O_1, \dots, O_k)$, regardless of body-completion order. An exclusive call overlaps no other tool body; cancellation drains started bodies and supplies synthetic outcomes for unstarted calls.

- Prepare and gate in assistant order; dispatch parallel-safe bodies in a bounded rolling pool; commit only the ready contiguous prefix.
- Five distinguishable rejections: parse, schema, policy, no usable confinement backend, and an admitted-but-failing execution.

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Recording the first rejecting owner keeps a policy denial from looking like a model-format error.

Guarded seams fail closed

- Ordered protocol: pre-execute policy, approval when requested, monotonic guards, wrapped provider dispatch, post-execute policy, result materialization.
- Missing approval denies an approval-dependent call. An unavailable sandbox backend refuses to run a command unconfined.
- A sandbox escalation must be strictly wider than the call's current mode, applies to that call only, and is approved before any effect.
- A post-rc.5 correction added a private PID namespace and private `/proc`: retaining the host PID namespace let a confined process follow `/proc/<pid>/root` into a host-visible mount view. [INSPECTED: pinned DSH snapshot, Bubblewrap correction commit]

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Fail-closed selection and backend completeness are separate claims. These filesystem backends do not by themselves restrict network access.

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[INSPECTED: pinned dsh-v0.1.1-rc.2 snapshot]

Visible stream progress is not automatically durable history. Recovery reconstructs from the log and immutable attachments — never from the broken stream.

Reasoning effort moves the operating point

Agent evaluation	Native metric	Non-think	High	Max
Terminal Bench 2.0	score	59.1	63.3	67.9

[REPORTED: DeepSeek-V4 technical report, V4-Pro]

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Seven of eight High/Max pairs improve at Max; MCPAtlas declines. So “always use Max” is not a supported per-task rule.

Concept-to-code map for the chapter

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Propose in the model. Authorize, confine, and remember in the application.

Next: the runtime chapter owns cache state, speculative decoding, and production serving; the training chapter owns how the Non-think, High, and Max policies are produced.